

Moveit Positive Kinematics Design

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1. Operating Environment

Ubuntu 22.04

ROS2: Humble

2. Driving the Real Machine

Driving the real machine involves subscribing to Moveit2's `/joint_states` topic to convert the joint state information of the robotic arm into control data for the actual robotic arm.

Note: Since the real robotic arm does not have obstacle avoidance functionality, some poses may encounter obstacles; therefore, the planned robotic arm movements should be as reasonable as possible and avoid locations with obstacles.

(It is recommended to use preset poses for the demonstration of driving the real machine)

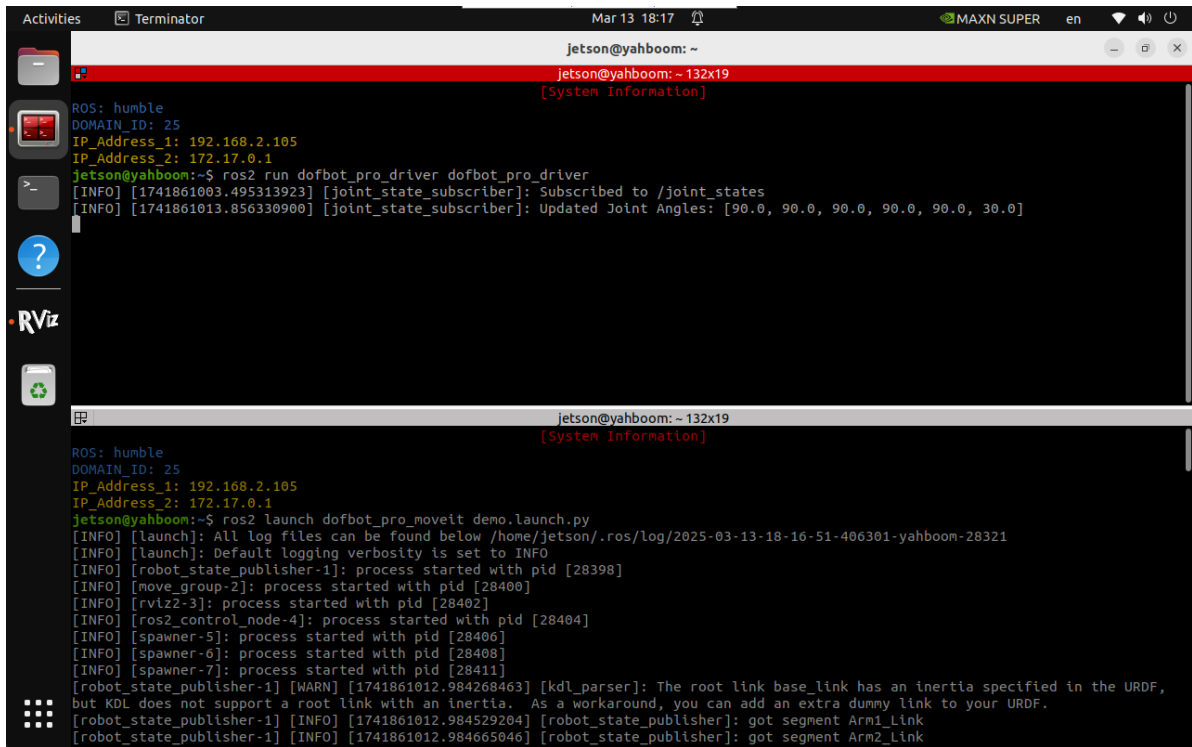
2.1 Starting the Real Machine

If not driving the real machine, demonstrate the robotic arm movements in MoveIt simulation (skip the following steps):

```
ros2 run dofbot_driver dofbot_driver
```

2.2 Starting MoveIt2

```
ros2 launch dofbot_pro_moveit demo.launch.py
```



```
jetson@yahboom: ~  
jetson@yahboom: ~ 132x19  
[System Information]  
  
ROS: humble  
DOMAIN_ID: 25  
IP_Address_1: 192.168.2.105  
IP_Address_2: 172.17.0.1  
jetson@yahboom:~$ ros2 run dofbot_pro_driver dofbot_pro_driver  
[INFO] [1741861003.495313923] [joint_state_subscriber]: Subscribed to /joint_states  
[INFO] [1741861013.856330900] [joint_state_subscriber]: Updated Joint Angles: [90.0, 90.0, 90.0, 90.0, 90.0, 30.0]  
  
jetson@yahboom:~$ ros2 launch dofbot_pro_moveit demo.launch.py  
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2025-03-13-18-16-51-406301-yahboom-28321  
[INFO] [launch]: Default logging verbosity is set to INFO  
[INFO] [robot_state_publisher-1]: process started with pid [28398]  
[INFO] [move_group-2]: process started with pid [28400]  
[INFO] [rviz2-3]: process started with pid [28402]  
[INFO] [ros2_control_node-4]: process started with pid [28404]  
[INFO] [spawner-5]: process started with pid [28406]  
[INFO] [spawner-6]: process started with pid [28408]  
[INFO] [spawner-7]: process started with pid [28411]  
[robot_state_publisher-1] [WARN] [1741861012.984268463] [kdl_parser]: The root link base_link has an inertia specified in the URDF,  
but KDL does not support a root link with an inertia. As a workaround, you can add an extra dummy link to your URDF.  
[robot_state_publisher-1] [INFO] [1741861012.984529204] [robot_state_publisher]: got segment Arm1_Link  
[robot_state_publisher-1] [INFO] [1741861012.984665046] [robot_state_publisher]: got segment Arm2_Link
```

3. Forward Kinematics Design

Forward Kinematics (Forward Kinematics) Kinematics refers to a robotic arm given a target pose (angles of each joint), which MoveIt then automatically plans to achieve.

Startup Command

After the robotic arm has successfully loaded MoveIt and displays `You can start planning now!`, run the following command: The robotic arm will automatically plan to the target pose.

```
ros2 run dofbot_moveit set_target_joints
```

